

Date : / /

Mon Tue Wed Thu Fri Sat Sun

Assignment No

Name: M. Ahmed Javed

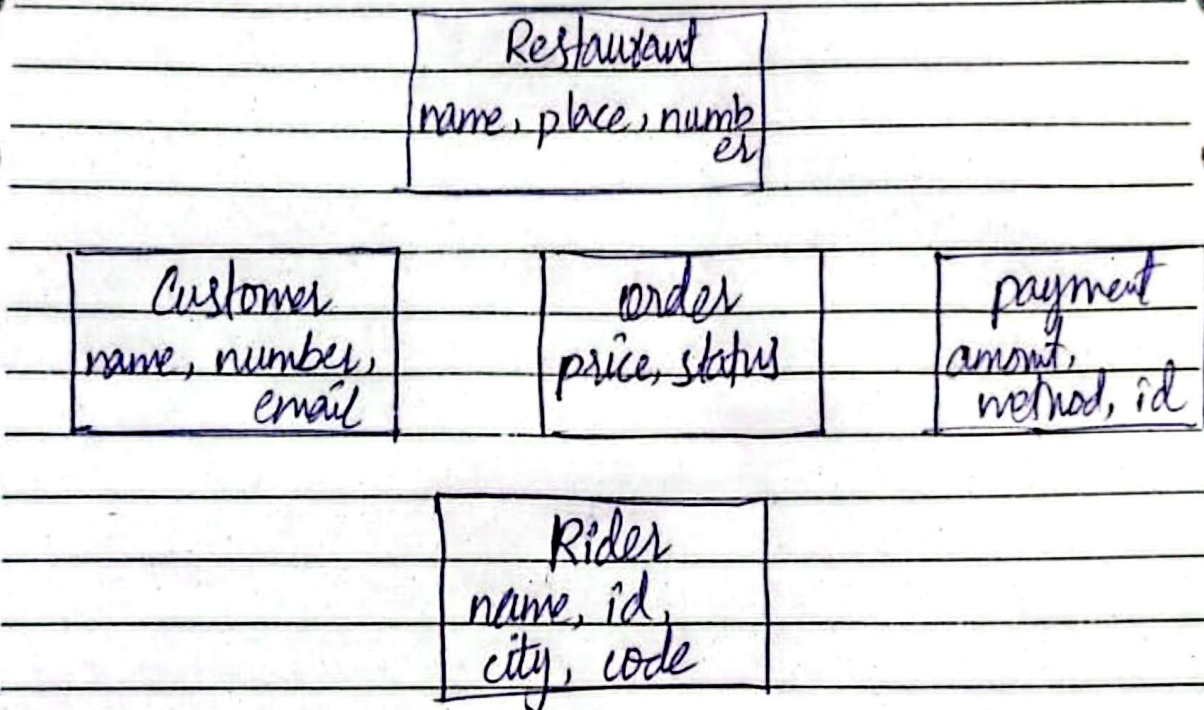
ID: 1167-2024

Sub: ADBMS

Sem: 04

Assig: 02

Q1: Diagram



Entities and Attributes:-

- ① Customer :- customer name, id, place, city
- ② Order :- name, id, status, city, total things
- ③ Restaurant :- rest name, place, city
- ④ payment :- payment method, status, amount
- ⑤ rider :- rider name, status, id



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6) Relational Method:-

The data is stored in the form of tables & relations. Each table consists of rows and columns.

Data representation:-

- ① Customer :- customer id, name, number etc
- ② Restaurant :- Restaurant name, id, place, etc

Table

Customer ID	Customer name	Restaurant name	Restaurant ID	order ID
1	Ahmed	KFC	5	32
2	Babar	Foxa	6	33
3	Adam	Hungers	7	35

Rider ID

073

074

075



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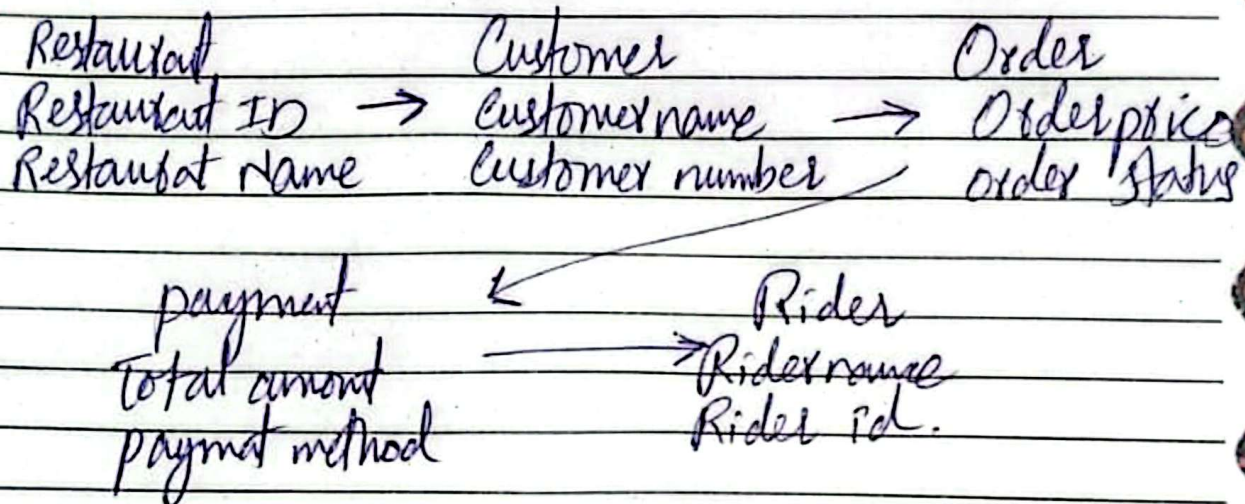
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Object Oriented Model:-

Data will be stored in the form of objects instead of table.

Representation:-



(c) Feature	Relational Model	Object Oriented Model
① Flexibility	Fixed schema less flexible	More Flexible support object
② Data sharing	Uses table and join slower in distributed system	shared data through objects more efficient in complex system
③ Complex relationship	Handled using multiple tables	Handles easy using object.



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Assignment No

(d) Relational Model:-

Fragmentation:- Table like Restaurant, order, delivery orders are divided by city.

Replication:- customers and payment are copied

Object oriented Model:-

Fragmentation:- Each cities maintains local object such as restaurant order

Replication:- customers and payment are replicated across cities

(e) Mongo DB is the good example of a Distributed database system because it supports.

① Distributed storage through sharing

② Data availability through replication

③ easy scaling across multiple cities or different places.



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(f) No SQL improve distributed data handling by making systems more scalable, flexible, and efficient.

Document Database are better for flexible multi-city storage.

Graph database are better for complex connected relationships.

(g) Multi Dimensional Data Model:-

It organizes data for analysis using dimension and measures.

Dimension $\rightarrow$ perspective for analysis

Measures $\rightarrow$ numeric values to analyze.

Associative Data Model:- It stores data based on relationships in between entities.

Example:- Restaurant receives order.

Context Data Model:- It stores data along with its surrounding conditions.

Example:- A delivery record includes order ID, location etc.

Spatio Temporal Data Model:- It stores

the data based on location and times.

Example:- Track a rider's location with timestamp during delivery.



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